Image Matching Based on Two-Column Histogram Hashing and Improved RANSAC

Bin Li, Delie Ming, Wenwen Yan, Xiao Sun, Tian Tian, and Jinwen Tian

***Abstract*—To improve computational efficiency in synthetic aperture radar (SAR) image matching, a fast image matching method using a novel two-step searching strategy (coarse-to-fine) is proposed in this letter. This method is based on two-column histogram (TCH) hashing and improved random sample consensus (RANSAC). First, coarse matching is conducted using a novel TCH hashing, which is notable for its robustness and speed. Compared with the discrete cosine transform used in perceptual hashing, TCH describes SAR images more accurately and rapidly. Then, in the refining stage, key points are detected and described in the coarser scales using scale-invariant feature transform. The Euclidean distance strategy and the improved RANSAC based on prior energy function (P-RANSAC) are then employed to implement matching. On the basis of prior information, a model of energy function has been constructed to improve sampling strategy. Experimental results on various SAR images show that the proposed approach outperforms the state-of-the-art algorithms in SAR image matching.**

***Index Terms*—Fast image matching, random sample consensus (RANSAC), scale-invariant feature transform (SIFT), two-column histogram (TCH) hashing.**

I. Introduction

IMAGE matching is a crucial step in many image analysis tasks, such as image registration, image fusion, object recognition environmental surveillance, etc. Among local features, scale-invariant feature transform (SIFT) [1] has become one of the most impressive methods for image matching and object recognition. Owing to the scale invariance of the detector and the distinctiveness of the descriptor, SIFT-based methods have been successfully used in remote sensing image matching or registration [2]–[7]. Compared with template matching, feature-based matching, such as SIFT, has higher precision but lower computational efficiency.

Perceptual image hashing functions produce hashing values based on the visual appearance of the image. Perceptual hashing can be also referred to as a robust hashing or a fingerprint [8]. A robust image hashing is applied for indexing and database searching by using the statistical property of wavelet coefficients [9]. Yang *et al.* [10] proposed a block mean value based image perceptual hashing algorithm. Fridrich and Goljan [11] introduced a robust hashing method whose hashing vectors are created by projections of discrete cosine transform (DCT) coefficients to key-dependent random patterns. Xiang *et al.* [12] proposed a robust image hashing algorithm by using the invariance of the image histogram shape to geometric deformations.

Template matching is one of the most common techniques used in synthetic aperture radar (SAR) image matching. Based on either the Euclidean distance or the cross correlation [13], the template matching algorithm finds the image patch that most closely matches the template image [14]. The coarse-to-fine strategy proposed by Rosenfeld and VanderBrug [15], [16] is a well-known approach to reduce the computational cost of template matching.

In certain circumstances, unmanned aerial vehicles fly in a complex electromagnetic environment, and we cannot directly acquire their precise positions. In addition, if we use the scene matching navigation, the sizes of two images significantly differ. Furthermore, the scale and orientation of these two images are basically consistent. We define the smaller image as the real-time image and use the larger image as reference. Image matching simply involves finding the matching points and matching position of the real-time image on the reference image. If we directly use feature matching, the matching speed and precision would be considerably decreased because of the large scope of the reference map search.

Combining the advantages of speed in template matching and precision in feature-based matching, we propose a novel coarse-to-fine matching method based on two-column histogram (TCH) hashing and improved random sample consensus (RANSAC) to solve this problem. We select local reference image using coarse matching and then reduce the search area of the fine matching on the reference image. The rest of this letter is organized as follows. The proposed TCH hashing and improved RANSAC based on prior energy function (P-RANSAC) are discussed in Section II. The experimental results are provided in Section III, followed by the conclusions in Section IV.

II. Proposed TCH Hashing and P-RANSAC

A. Coarse Matching: TCH Hashing

Although both mean hashing [10] and DCT hashing [11] can depict the image contour information excellently, these

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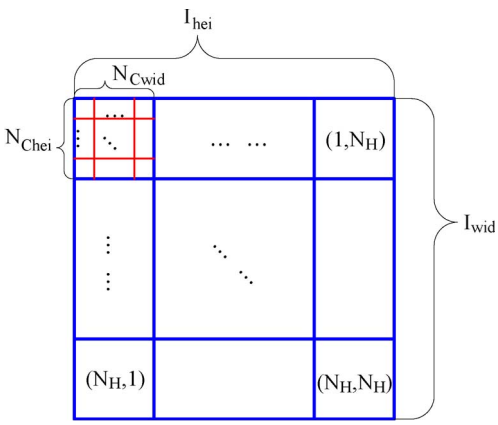

Fig. 1. Structure of image cells.

methods are easily affected by noise because of their poor robustness. Considering the simplicity and efficiency of gray difference, as well as the distinction of SAR image speckle noise, together with inspiration from local binary patterns [17] and contrast context histogram [18], we propose a perceptual hashing feature that can effectively restrain the noise.

In the process of constructing TCH hashing features, one of the core steps is producing the TCH. The term TCH [19] is defined as a projection of the statistical histogram for all the subblock pixels by using a certain mapping relationship. In our algorithm, we define the subblock of perceptual hashing as a cell of the image. As shown in Fig. 1, the image is first divided into $N_H \times N_H$ cells. The size of cells is $N_{C\mathrm{hei}} \times N_{C\mathrm{wid}}$ pixels ($N_{C\mathrm{hei}} = [\mathrm{Hei}_I/N_H]$, where Hei_I is the height of the image, and $N_{C\mathrm{wid}} = [\mathrm{Wid}_I/N_H]$, where Wid_I is the width of the image). We then calculate the mean $\overline{I}_{m,n}$ and variance $\sigma_{m,n}$ of pixels in each cell region ($m = 0, 1, \ldots, N_H$; $n = 0, 1, \ldots, N_H$). Taking $\overline{I}_{m,n}$ and $\sigma_{m,n}$ as the projection formula parameters, we compare the gray difference of each pixel with certain combinations of the mean and the variance, and then, finally, transform the pixels in each cell into a ternary value as $C_{m,n,i,j}$, $m = 0, 1, \ldots, N_H$, $n = 0, 1, \ldots, N_H$, $i = 0, 1, \ldots, N_{C\mathrm{hei}}$, and $j = 0, 1, \ldots, N_{C\mathrm{wid}}$. The formula for the ternary-valued pixels in the cell is as follows:

$$\begin{cases} C_{m,n,i,j} = 1, & I > \overline{I}_{m,n} + k_I \times \sigma_{m,n} \\ C_{m,n,i,j} = -1, & I < \overline{I}_{m,n} - k_I \times \sigma_{m,n} \\ C_{m,n,i,j} = 0, & \overline{I}_{m,n} - k_I \times \sigma_{m,n} < I < \overline{I}_{m,n} + k_I \times \sigma_{m,n} \end{cases} \tag{1}$$

where k_I is the gray variance coefficient.

After the ternary-valued process, we separately count the pixel numbers $\mathrm{Num}_{C=1}$ and $\mathrm{Num}_{C=-1}$ of $C_{m,n,i,j} = 1$ and $C_{m,n,i,j} = -1$ to build the TCH of the cell. Then, each cell obtains a 2-D feature of the statistical histogram of the gray difference, i.e.,

$$\begin{cases} \mathrm{Num}_{m,n,C=1} = \underset{i=1,j=1}{\overset{N_{C\mathrm{hei}},N_{C\mathrm{wid}}}{\mathrm{Sum}}} (C_{m,n,i,j} = 1) \\ \mathrm{Num}_{m,n,C=-1} = \underset{i=1,j=1}{\overset{N_{C\mathrm{hei}},N_{C\mathrm{wid}}}{\mathrm{Sum}}} (C_{m,n,i,j} = -1). \end{cases} \tag{2}$$

We can obtain the feature vectors with a dimension of $N_H \times N_H \times 2$ based on the TCH of each cell calculated.

The essence of the histogram hashing algorithm is as follows. First, we separately calculate the mean and the variance of the $N_H \times N_H$-dimensional cell feature vectors of $\mathrm{Num}_{m,n,C=1}$ and $\mathrm{Num}_{m,n,C=-1}$. Then, we use a similar strategy as cell ternary-valued pixels and obtain the ternary-valued hashing $H_{i,C}$ $(i = 0, 1, \ldots, N_H \times N_H;\ C = 1,\ -1)$ using the dynamic variance threshold, which is different from the previous binary hashing. This way, $N_H \times N_H \times 2$-dimensional histogram ternary-valued hashing can build more stable and distinct image descriptions by adjusting the threshold. Experiments and analysis show that this method robustly performs against the speckle noise of the SAR images, i.e.,

$$\begin{cases} H_{i,C} = 1, & \mathrm{Num}_{m,n,C} > \overline{\mathrm{Num}_C} + k_{\mathrm{Num}} \times \sigma_C \\ H_{i,C} = 0, & \overline{\mathrm{Num}_C} - k_{\mathrm{Num}} \times \sigma_C < \mathrm{Num}_{m,n,C} \\ & < \overline{\mathrm{Num}_C} + k_{\mathrm{Num}} \times \sigma_C \\ H_{i,C} = -1, & \mathrm{Num}_{m,n,C} < \overline{\mathrm{Num}_C} - k_{\mathrm{Num}} \times \sigma_C \end{cases} \tag{3}$$

where $\overline{\mathrm{Num}_C}$ is the TCH mean of the image cells, σ_C is the variance, and k_{Num} is the variance coefficient of the TCH.

For coarse matching, the entire reference image is processed by successively extracting subimages from it. The sizes of the subimages are the same as the size of their real-time image. Then, we calculate the TCH hashing features of both the real-time image and the subimages. We use the Hamming distance as the similarity measure for the TCH hashing features of the real-time image and each subimage and find the most similar subimage that has the minimum Hamming distance. The position of the subimage is the initial position of the real-time image in the reference image. For the details on Hamming distance, we refer to the work of Swaminathan *et al.* [20].

B. Fine Matching: SIFT + P-RANSAC

SIFT is selected for multisource SAR image fine matching. Given the effect of the speckle noise in SAR images, numerous mismatching points may exist, which can significantly reduce the execution efficiency of RANSAC. To solve this problem, an energy function model is proposed to measure the similarity of feature points.

As a precondition for using RANSAC, a certain rule is observed for the correct positions of matching pairs among the SIFT points. This rule enables us to use the affine transformation model for the mapping. For correct matching points, the distributions of scales and principal directions of the correct matching points show a similarity and a small Euclidean distance. By contrast, without any regulation to follow, these distributions of scales and directions remarkably differ. In the scene matching navigation of SAR images, using the inertial navigation and parameters of SAR platform results in a consistent resolution of real-time and reference images; meanwhile, only a tiny rotation angle between them appears. According to the principle discussed in the preceding paragraphs, the correct matching points satisfy

$$|\theta_{\mathrm{rt}} - \theta_{\mathrm{ref}}| < \tau_\theta \tag{4}$$

$$|\sigma_{\mathrm{rt}} - \sigma_{\mathrm{ref}}| < \tau_\sigma \tag{5}$$

where $(\sigma_{\mathrm{rt}}, \theta_{\mathrm{rt}})$ and $(\sigma_{\mathrm{ref}}, \theta_{\mathrm{ref}})$ are the scales and the principal directions of the real-time and reference images, respectively.

Based on the aforementioned characteristic, we build the similarity energy function as follows:

$$E(n) = d_r(n) \times (\lambda \times d_\theta(n) + (1-\lambda) \times d_\sigma(n)) \quad (6)$$

where

$$d_r(m) = \mathrm{dec}(m) + \delta \quad (7)$$

$$d_\theta(m) = |\theta_{\mathrm{rt}}(m) - \theta_{\mathrm{ref}}(m)| \quad (8)$$

$$d_\sigma(m) = |(\sigma_{\mathrm{rt}}(m) - \sigma_{\mathrm{ref}}(m)) / (\sigma_{\mathrm{rt}}(m) + \sigma_{\mathrm{ref}}(m))| \quad (9)$$

where dec denotes the Euclidean distance of the feature points, and δ denotes the similarity suppressive coefficient of the Euclidean distance. The contribution of dec to the energy function can be reduced or increased by changing δ. λ is the scaling factor of the measurement factor, and it can change the weights d_θ and d_σ. d_r is the punishment factor of distance, d_θ is the punishment factor of principal direction, and d_σ is the punishment factor of scale. As the value of E decreases, the similarity between two feature points and the possibility of being a correct matching pair increase. We rearrange the feature points based on the value of E and use the top $1/N$ of the matches to perform RANSAC. Upon detecting the result and the model, we implement the consistency test to all of the feature points and then calculate the correct matching points.

III. Experiments and Results

Here, we demonstrate the performance of our method using a large test set that contains 342 test image pairs. The real-time image is the airborne SAR data of an unmanned aerial vehicle with the size of 400 × 400 pixels. The reference image is the satellite-based SAR data of RADARSAT II with the size of 1000 × 1000 pixels. Both the real-time and reference images are ground-range gray scale images. The software platform of the algorithm is MATLAB2009A, and the executive program is written in MATLAB codes or hybrid codes of MATLAB and C. The central processing unit of the running machine is Core i5, with a frequency of 3.2 GHz.

A. Algorithm Process

The entire coarse-to-fine matching procedure can be described as follows.

Step 1) Preprocess the real-time SAR image and the reference SAR images (by mean filtering and histogram equalization).

Step 2) Process the whole reference image by successively extracting the subimages; then, calculate the TCH hashing features of both the real-time image and the subimages and use the Hamming distance [26] as similarity measurement to locate the initial position of the real-time image in the reference image.

Step 3) Based on the result of the coarse matching (Step 2), the subreference image with the size of 450 × 450 will be extracted to perform the fine matching.

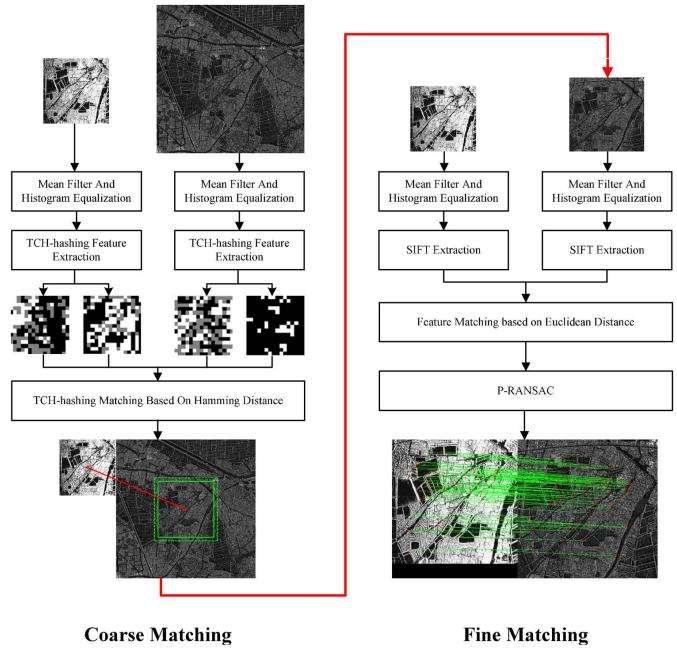

Fig. 2. Flowchart of the proposed approach.

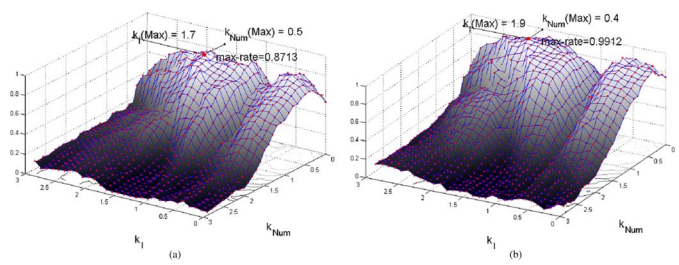

Fig. 3. Parameter analysis for TCH hashing matching rate. (a) Cell sized 8 × 8. (b) Cell sized 16 × 16.

Step 4) Use SIFT to detect and describe the feature points, and use the Euclidean distance to measure similarity.

Step 5) Use the improved P-RANSAC algorithm to remove the mismatching points calculated in Step 4. Then, the final matching point pairs can be outputted to generate a transformation model.

The flowchart of the proposed approach is illustrated in Fig. 2.

B. Coarse Matching Experiment

In the TCH hashing matching, both values of the gray variance coefficient k_I and the TCH variance coefficient k_{Num} directly affect the experimental results of the coarse matching. In Fig. 3(a), the 8 × 8 cell is used in the calculation of the TCH hashing feature of the real-time image. The size of each cell is 50 × 50 pixels, and the dimension of the final feature is 8 × 8 × 2. With the value of $k_I = 0.5$ and $k_{\mathrm{Num}} = 1.7$, we obtain 298 correct matches for a total of 342 SAR images and achieve an 87.13% accuracy rate. In this letter, if the error of the matching position is less than 25 pixels, we consider the coarse matching result as correct. In Fig. 3(b), the 16 × 16 cell is used in calculating the TCH hashing feature of the real-time image. The size of each cell is 25 × 25 pixels, and the final feature

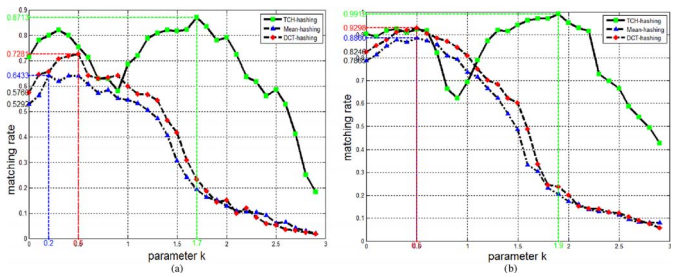

Fig. 4. Parameter analysis for TCH hashing, mean hashing, and DCT hashing matching rates. (a) Cell sized 8 × 8. (b) Cell sized 16 × 16.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MATCHING METHODS

	8*8 cell		16*16 cell	
	Matching rate	k	Matching rate	k
Original mean hashing	0.5292	0	0.7865	0
Original DCT hashing	0.5760	0	0.8246	0
TCH hashing	0.7164	0	0.8860	0
Improved mean hashing	0.6433	0.2	0.8860	0.5
Improved DCT hashing	0.7281	0.5	0.9298	0.5
Optimal TCH hashing	0.8713	1.7	0.9912	1.9

is 16 × 16 × 2 dimensional. With the value of $k_I = 0.4$ and $k_{\mathrm{Num}} = 1.9$, we obtain 339 correct matches for the same test set and achieve a 99.12% accuracy rate.

After the testing of TCH hashing parameters, we compare the three hashing techniques, namely, mean, DCT, and TCH, using 8 × 8 and 16 × 16 cells. We also test our improved mean hashing and DCT hashing algorithms using a ternary hashing value similar to the one produced in the TCH hashing algorithm [refer to formula (6)]. With a variable k (in TCH hashing, $k = k_{\mathrm{Num}}$) ranging from 0 to 2.9 with an interval of 0.1, we use all of the 342 SAR image pairs in our experiments, where the gray variance coefficient k_I of TCH hashing is 0.5 for the 8 × 8 cell and 0.4 for the 16 × 16 cell.

Fig. 4(a) and (b) show the coarse matching experimental results in different cells. The case of $k = 0$ indicates a traditional binary perceptual hashing algorithm. The experimental results on the optimum matching rate are presented in Table I. The matching probability of the original mean hashing is the lowest and followed by that of the DCT hashing. The matching rate for our algorithm is significantly higher than the other algorithms listed in Table I.

In the next section, the time efficiency of each algorithm is determined and compared with one another. Fig. 5 presents the required matching time to complete the coarse matching procedure once on the average of three algorithms. The cell sizes are 8 × 8 and 16 × 16, and the searching step of the reference image is 25 pixels. The experimental results indicate that the execution speed of the mean hashing is the highest at 0.0729 s (for the 16 × 16 cell), and that of the TCH hashing is 0.1541 s. The execution speed of the DCT hashing, which exceeds 0.5 s, is the worst.

Basing from the comparative experiment of the aforementioned matching algorithms, we can conclude that the matching probability of the TCH hashing algorithm is the highest. Its time efficiency is also significantly high. Although the matching time of the TCH hashing algorithm is slightly longer than that of the mean hashing algorithm, it is considerably shorter than that of the DCT hashing algorithm. Thus, we select the 16 × 16 cell for the TCH hashing matching algorithm used as the coarse algorithm in this letter.

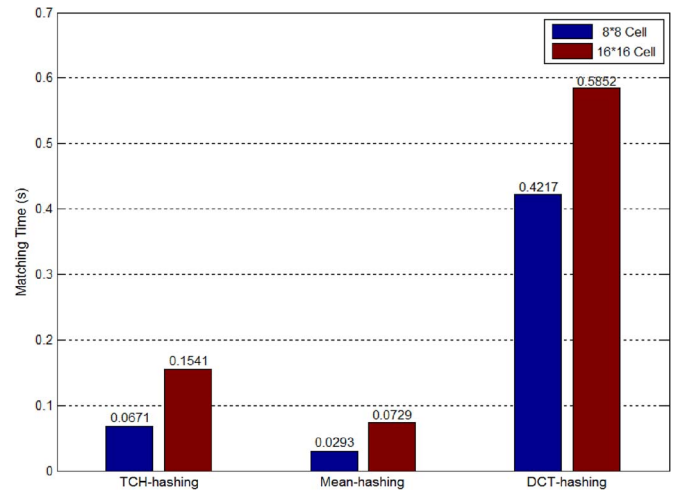

Fig. 5. Matching time analysis for TCH hashing, mean hashing, and DCT hashing matching.

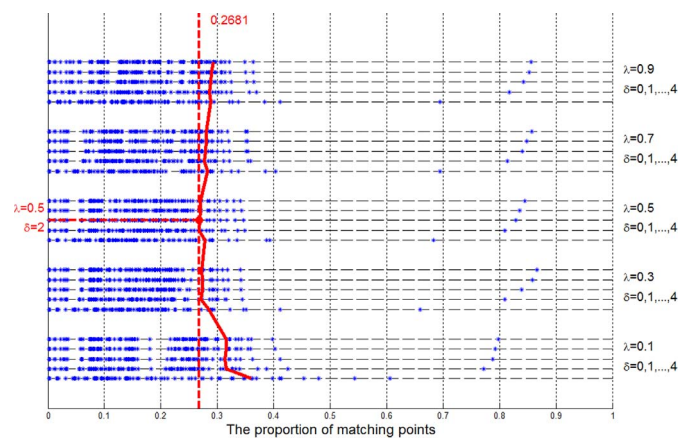

Fig. 6. Parameter analysis for P-RANSAC.

C. Fine Matching Experiment

Before the fine matching, we estimate the three parameters of the energy function $E(n)$, namely, δ, λ, and N, in the P-RANSAC algorithm by conducting experiments.

As Fig. 6 shows, the different values of the parameters δ and λ affect the distribution of the correct matching points. All matching points are sorted from small to large using function $E(n)$, and the blue points represent the correct matching points. The distribution range is from 0 to 1. The red solid line represents the boundary that contains the 90% correct points. Fig. 6 shows that the energy value of most of the matching points is considerably smaller than the mismatching points. $E(n)$ has a more distinct characteristic for the matching points, and it is not sensitive to the parameters. Basing from the experimental results, we then select the best parameters: $\delta = 2$, $\lambda = 0.5$, $N = 3$.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MATCHING METHODS

Matching algorithm	C-matching time (s)	F-matching time (s)		Total time (s)	Correct \ Match points
		Sift	RANSAC		
SIFT+RANSAC	——	4.51	122.90	127.41	49/1088
SIFT+P-RANSAC	——	4.49	3.00	7.49	49/1088
TCH-hashing+ SIFT+RANSAC	0.1541	1.47	40.42	42.04	68/1088
TCH-hashing+ SIFT+P-RANSAC	0.1541	1.48	1.07	2.70	69/1088

Based on the result of the coarse matching, a fine matching image sized 450×450 is intercepted in its corresponding position in the 342 reference images. We conduct the fine matching experiment by using the improved method of removing the mismatching points and then conduct a comparative experiment with the original algorithm.

Table II compares the dissipative time, SIFT matching point pairs, and correct point pairs after we remove the mismatching points in every part of the four algorithms. All of the results indicate the average values of the 342 pairs of images. The first line illustrates the original unimproved SIFT algorithm without the coarse matching. Although the time of the coarse matching is not observed, the times of both the SIFT matching and the RANSAC are considerably long. The second line presents the results with the improved P-RANSAC but without the coarse matching. Thus, the removal time for the mismatching points is largely reduced. The third line shows the results with the coarse matching and the unimproved RANSAC. The time for SIFT matching is reduced by one third, and the number of correct matching points is markedly increased. The fourth line shows the results with the coarse matching and the improved P-RANSAC. The overall time of the algorithm is significantly reduced, and the number of correct matching points is the maximum.

IV. CONCLUSION

In this letter, a novel coarse-to-fine fast image matching method has been proposed. To improve the speed and probability of matching, we use a rapid template matching based on TCH hashing in the coarse matching. Then, an improved P-RANSAC algorithm is applied to remove the mismatching points for better precision. The coarse matching experiment compares the expressive power of the template matching for the SAR image between the traditional hashing and the TCH hashing algorithms. The fine matching experiment demonstrates the execution efficiency of the proposed algorithm. The experimental result validates that the TCH hashing algorithm effectively suppresses the noise of the SAR image. Moreover, the TCH hashing technique describes the image more accurately and achieves a higher real-time performance.

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